

SandSational Plan

Yes, for a browser game like Particul, I would not use only React.

Particul is an incremental/idler game with click-to-mine particles, particles falling to the bottom, automation, selling, upgrades, research, trading, and rare particles. Steam tags it as Incremental, Idler, Physics, Mining, Pixel Graphics, Automation, 2D.

For a browser version with a mandatory Java backend, I'd use:

Stack

Frontend

Phaser 3 + TypeScript + Vite

UI Overlay

React optional, but not required at the start

Backend

Java Spring Boot

Database

PostgreSQL

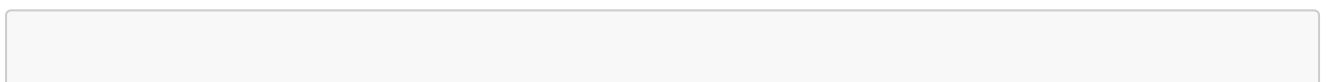
Best Frontend Choice: Phaser

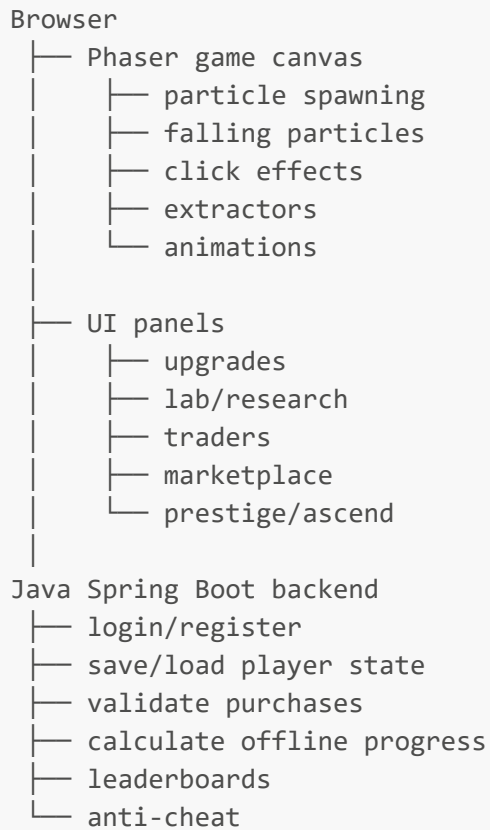
For this specific style, Phaser is the right frontend/game framework because you need:

- particles falling
- click mining
- 2D physics-like movement
- sprites
- animations
- object pooling
- floating numbers
- game loop
- effects

React alone is not ideal for thousands of moving particles. React is better for menus and panels, not rendering a physics-like particle pile.

Recommended Architecture





Communication: REST First, Not WebSocket First

For a Particul-like game, I would still start with REST.

Use REST for:

```
GET  /api/game/state
POST /api/game/click
POST /api/game/buy-extractor
POST /api/game/sell-particles
POST /api/game/research
POST /api/game/trade
POST /api/game/ascend
```

Use WebSocket only later for:

- live leaderboard
- global events
- chat
- market price changes
- multiplayer trading
- server announcements

Important Design Rule

The browser should handle the visual particles, but the backend should handle the real economy.

Good design:

Phaser Shows

- particles falling
- animations
- estimated income
- nice effects

Spring Boot Calculates

- real particle count
- upgrades owned
- production per second
- offline gains
- purchase validation

Do not let the frontend say:

```
"I earned 999999 particles"
```

Instead, frontend says:

```
"I clicked"  
"I want to buy upgrade"  
"I want to sell particles"
```

Then the Java backend calculates whether that is valid.

My Final Recommendation

For a browser game inspired by Particul:

- Phaser 3 + TypeScript + Vite
- Java Spring Boot
- PostgreSQL
- REST API first
- WebSocket later only if needed

React can be added later for complex menus, but I would start with Phaser + TypeScript first.

Resourecs

- Sand

- Dirt
- Stone
- Silver
- Gold
- Platinum
- Emerald
- Diamond
- Obsidian